A Case Study on User Perception of Parameterized LLM-Generated Narratives

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***Abstract*—Storylet-based narrative systems offer a flexible structure for interactive storytelling, but authoring a diverse set of meaningful storylets requires significant design effort. Large language models (LLMs) may have the potential to alleviate this burden by generating short-form narratives in response to predefined structural constraints for use in such systems. In this paper, we conduct a case study on ChatGPT 4o's ability to generate narrative fragments with parameterized intensity levels, and assess the alignment of user perceptions with these intended intensities. We hypothesize that newer large language models such as GPT-4o are able to effectively generate short-form narratives in response to parameterized inputs, and that user assessment of these LLMs' output will largely be in line with their intended intensities. To test this, we generate fifty stories across five unique story domains, each varying in a single parameter relevant to the story, such as success, damage to a relationship, chaos caused, or others, on a scale of 1 to 10. In a user study, participants were asked to compare pairs of stories with varying intensity differences (1, 2, and 3) within each story domain. Analysis of user feedback, gathered through Likert-scale surveys, indicates a strong correlation between user-perceived intensity and the intended intensity ordering of generated stories. Additionally, user certainty in comparisons increased as the intensity gap between story pairs increased. The results support our hypothesis that LLMs can effectively generate stories with discernible intensity variations in response to parameterized input, and that users can reliably perceive these variations as intended. This research contributes to understanding the potential of LLMs in controlled narrative generation and provides support for their use in LLM-aided, hybrid narrative generation systems or games in the future.**

I. Introduction

Large language models (LLMs) such as GPT-3 and GPT-4o [1] are proving to be powerful tools for approaching text-based tasks, easing authorial burden and barriers to entry in various areas. In the field of interactive narrative games and story generation systems, one of the more laborious tasks is the generation of plot fragments or "storylets": atomic units of narrative in the context of a storytelling game or system, with accompanying preconditions, postconditions, and associated story content [2].

Though not ubiquitous within storytelling games, storylet-based systems allow for a narrative to be forged during play by cobbling together sequences of these narrative chunks, using their preconditions to ensure only storylets appropriate for a given story context occur at any given point. Each storylet, in turn, alters the story world as it occurs, pushing the narrative forwards. In this way, narratives dynamically unfold based on a player's actions and the game state, allowing for sequences of events to occur in interesting, but not explicitly authored ways [3]. Such systems have seen success within works such as Failbetter Games' interactive fiction / text adventure *Fallen London* [4] or the *Sunless Sea* RPG, where they empower players to shape their own story paths.

However, authoring storylets presents a significant challenge, as each must be carefully crafted to fit within the broader system. Appropriate preconditions and postconditions must be authored for each storylet, ensuring they trigger in appropriate contexts and produce relevant consequences for the stories being told. Beyond this, one *still* must author the actual narrative content or discourse, which similarly must be appropriately fitted to the preconditions and postconditions of the storylet it belongs to.

This challenge ties directly into the concept of authorial leverage—the power a tool gives an author to define a quality interactive experience in line with their goals, relative to the tool's authorial complexity. [5]. Storylets provide a powerful structure for enabling responsive, player-driven storytelling, but their implementation places a high burden on authors to manually craft numerous story fragments [3]. LLMs present a potential solution by automating the generation of storylet discourse—the natural language descriptions of events unfolding within the structured scaffolding of a storylet system. If LLMs can reliably generate coherent and contextually appropriate narrative fragments, they could significantly increase authorial leverage by allowing designers to focus on defining structural constraints rather than writing every piece of content manually.

However, while there is potential to speed up the development cycle of story-driven narrative games by utilizing LLMs, LLMs have previously fallen short of expectations for full story generation, encountering issues in constructing satisfying story arcs, managing reader experience, and maintaining narrative consistency among others [6]. LLMs struggle to construct coherent long-form narratives, often losing track of prior events and generating inconsistencies that break reader immersion in the process [7].

Despite these limitations, LLMs may be more effective within a storylet-based framework, where they are responsible for generating only short-form narrative discourse rather than full-length stories. A storylet system inherently provides a logical backbone—preconditions and postconditions—which can be used to constrain and guide how the LLM-generates story text. Instead of requiring LLMs to manage long-term narrative progression, this approach allows them to focus on local coherence within individual storylets, reducing the risk of hallucinated plot points or contradictions. By leveraging curated prompt designs and storylet scaffolding, we hypothesize that LLM-generated storylet discourse can remain contextually appropriate while still benefiting from the efficiency and expressivity of machine-generated prose.

This paper contributes a closer examination and an evaluation of the ability of LLM-generated narrative fragments to meet an author's intentionality, by conducting a user study upon generated narrative fragments to assess the alignment of user perceptions with author intentions. Our study measures participant agreement with the expected intensity ordering of pairs of narrative fragments, providing insight into whether LLMs can be effectively harnessed to enhance authorial leverage in storylet-based systems.

II. Related Works

To contextualize our work, we will discuss a few examples of storylet systems that we believe have the potential to utilize LLM text generation in order to increase authorial leverage, as well as examine existing LLM-aided story generation systems and discuss their shortcomings. Lastly, we will discuss a number of recent approaches in evaluating stories generated by LLMs, to justify our user-study methodology for evaluating the capabilities of LLMs to respond to parameterized inputs.

A. Storylet Systems

While not the first narrative game to use a storylet-like methodology to drive story progression *Fallen London* [4] is the first game to employ the term "storylet" [8] explicitly. Emily Short provides a concrete definition of a storylet [2] as a self-contained unit of narrative content that includes:

1. A segment of narrative such as dialogue, narration, etc.
2. Prerequisites that determine when the narrative content can play.
3. Postconditions that alter the world state as a result of the narrative content.

In the context of a text-based narrative adventure game like *Fallen London*, storylets primarily operate as avenues for players to choose how to progress further in the game. Storylets are unlocked once the player's attributes meet their prerequisites, and success in storylet events will generally advance the story and provide a boon to the player's stats. For example, in the beginning of Fallen London, the player is confronted with a need to escape a prison. By selecting a storylet in which your character attempts to lift a tool off a guard to pry the bars open, the player is greeted with a text-based description of their actions and success or failure, with success leading to adjusting the player's stats, unlocking additional storylets that allow for the jailbreak to continue. While this approach is highly effective at ensuring salient story text and events are presented to the player, there is significant overhead in the bookkeeping required. Investigating methodologies that would allow for building these storylets with less authorial burden would allow for the same amount of development resources to yield a more dense narrative space, allowing for more user choice and unique experiences.

Lume [9] is another interesting storylet-based approach to procedural story generation. Focusing on improving narrative coherence, Lume builds additional layers of details into the structure of these storylet objects. Now making room for bindings that pull other non-player characters into a particular storylet to interact with the player, and optional *recall phrases* that allow for storylets to refer back to triggering events, it helps weave a greater sense of causality into the stories it produces. With this, however, is greater still incurred cost upon those who wish to author for the system. While Lume shows great potential for narrative coherence, it also requires additional design effort to ensure that rules, bindings, and control flow uphold that coherence. As before, we see a growing need for tools to aid in reducing this authorial burden.

College Ruled [10] is an example of a system that leverages storylets in a different manner than a traditional text-based narrative game. College Ruled uses storylets (referred to as "plot fragments" in the context of this work) as a way to map out a space of possible stories that can be told from a given starting point. By comparing a worldstate—a specific configuration of the story world, characters, and their attributes—with a database of storylets and their preconditions, the system determines which storylets can run and which neighboring worldstates can be reached. By constructing a graph of narrative space by repeating this process, and pairing it with a search methodology that allows for authors to construct sets of story-waypoints, the system seeks to construct narratives that meet an author's goals. Such a system would similarly benefit from methods to accelerate the development of the storylets it uses, as the possible narratives the tool can be used to generate is heavily restricted by the library of plot fragments available to it.

Other recent endeavors have investigated pairing LLMs with storylet systems (referred to as "triggers" in the context of this work) explicitly. Drama Llama [11][1] leverages LLMs to allow for storylet preconditions and postconditions to be expressed in natural language, within an interactive narrative, and to generate character responses to events as they unfold. By using an LLM to determine if a storylet's preconditions have been met, Drama Llama allows for more flexibility than previous systems, which frequently rely on identifying complex combinations of tracked game state conditions to satisfy a storylet's preconditions before it can fire. Although the system uses LLMs to reduce authorial burden, it does so primarily through easing the process of configuring storylets for use, rather than in the generation of storylet text, as

[1] Drama Llama is not a peer-reviewed publication, but is still highly relevant recent work in this particular area of study.

we wish to investigate in this study. Furthermore, the work presents a preliminary evaluation of the tool's effectiveness, and merits a more thorough investigation into the abilities of an LLM to accurately trigger events in response to natural language—though the promise of simpler storylet authoring is greatly appealing.

B. Challenges in Long-Form LLM Storytelling

Utilizing LLMs for narrative generation tasks has, in the past, generated mixed results.

One such approach taken by Dueifi and Eger [7] investigated the capabilities of LLMs to generate narratives when used in conjunction with an existing narrative planning system, Glaive [12]. For context, Glaive focuses on abstract story representation, constructing a sequence of events that describe a story at a high level, with the aim of generating a story that meets an author's goals, while only being composed of sensible character actions, given character motivations. As it focuses solely on finding a valid narrative solution, however, the story representations it produces aren't able to be experienced in the same way that readable story text is. Dueifi and Eger propose a prompting methodology to use LLMs to expand the narrative plan produced by Glaive into long-form narratives (approximately 25,000 words). However, significant limitations were observed in the quality of the produced narratives. Despite efforts to alleviate context-window related issues inherent to transformer models by augmenting the LLM with a narrative planner. The system still struggled to keep a cohesive tone in its story generation. Problems ranged from turning what should have been a single character into a trio of protagonists, providing inaccurate information in the context of produced stories (suggesting ordering whiskey to sober up), and myriad other problems. Ultimately, the work provides interesting insight into the current issues facing long-form story generation tasks, and suggests that there may be further advancements needed before reliable novella-length output can be achieved with LLMs.

These challenges suggest that LLMs may be better suited to generating short, self-contained narrative fragments rather than full-length stories. A storylet framework, which inherently constrains the scope of each generated piece, may mitigate these issues by allowing LLMs to focus on local coherence rather than long-term narrative progression.

C. LLM-generated story evaluations

Here we will talk about the existing works focusing on evaluating LLM-generated stories. Recently, there has been a thrust to move towards automated evaluation of LLM-generated narratives. Part of the reasoning for this is mentioned in the previous work by Dueifi and Eger. When discussing evaluation methodologies for their LLM-generated narratives, they state: "the sheer quantity of text the readers would have to go through is beyond any reasonable compensation we could offer" [7]. Clearly, there is motivation to find alternatives due to the difficulties in sourcing human-based evaluations of large volumes of text.

In "Do Language Models Enjoy Their Own Stories? Prompting Large Language Models for Automatic Story Evaluation" [13], Chhun et al. investigate the capabilities of LLMs in evaluating LLM-generated narratives, comparing a number of automated methods such as assessed BLEU or ROUGE-1 scores alongside LLM-generated evaluations, and human evaluations. They investigate the correlation between generated scores, and find that while LLM ratings generally correlate similarly with other automated metrics or exceed them in performance, correlations of automated evaluations of stories with human evaluations of stories lags behind.

Shankar et al. attempt to align automated story evaluation methodologies with human methods in "Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences" [14], but still encounter issues. They identify that attempting to align automated evaluation methodologies with human evaluations is an important task for appropriately evaluating narrative works, but in their efforts to tune automated evaluation methodologies to user input, identify a phenomenon they dub "criteria drift". As participants graded more LLM outputs to assist in tuning, their criteria to evaluate those LLM outputs would be altered over time by their previous experiences. This poses a challenge to appropriately tuning automated LLM evaluations to human standards.

Given these challenges in generating automated evaluations for generated narratives, human evaluation remains the gold standard for this task at the present moment, despite its high cost. As a result, this study adopts a human-centered approach to evaluating LLM-generated storylet prose.

III. Methodology

To assess the capabilities of GPT-4o to generate short narratives in response to parameterized inputs, we conducted a user study in which 28 participating respondents were asked to evaluate a series of short stories generated on parameterized inputs.

A. Story Generation

To generate stories for comparison, we utilized OpenAI's GPT-4o model, specifically, the gpt-4o-2024-05-13 model snapshot. GPT-4o was repeatedly prompted and asked to provide 10 paragraph-long stories about a specific story domain, varying in a relevant domain-specific story parameter on a scale of 1-10. A brief description of what a "1" and a "10" on the aforementioned should entail was supplied, and a request was made to avoid explicitly referencing the scale description provided within generated stories. For example, one such prompt was the following: "Let's do another 10 instances of stories. Give me 10 short, paragraph-long story snippets about a character enjoying a brunch. The brunch should vary in quality on a scale of 1-10, from quite mediocre, to delectable. Avoid using the term "mediocre" or "delectable" in the descriptions".

This process was repeated until GPT-4o had generated fifty stories across five unique story domains, each varying in a single parameter relevant to the story, such as success, damage to a relationship, chaos caused or others, on a scale of 1 to 10.

This methodology was arrived at after some experimentation with the system—initial efforts at producing sets of stories tended to close each with an explicit statement about the results of the scene, in accordance with its place on the described scale. For example, a prompt regarding stories depicting damage to a relationship indicating that "...a 10 represents complete destruction of any friendship" generated a response for which the 10/10 scale storylet closed with: "Their friendship is completely destroyed, and they part ways with no intention of reconciling". To allow for the investigation to focus on the ability of users to identify the intensity of textual descriptions rather then simply match the final line of the generated story with the scale, prompts were adjusted accordingly. Otherwise, GPT-4o responded very well to prompting, always generating the appropriate amount of stories in response, and all generated stories were within the desired story domains.

Regarding story domains, we selected five intended to cover a breadth of different storylet archetypes to test performance across distinct dimensions of narrative generation. Two are centered around pairwise character interactions: "Flirting Success" and "Relationship Damage", but capture opposing relational trajectories. "Repair Success" and "Havoc Caused" both focus on agent–environment interactions, enabling assessment of how well models can generate outcomes based on task-oriented actions. Finally, "Brunch Quality" is intended to examine the LLMs' ability to describe experiences with distinguishable levels of pleasantness.

B. Survey Details

With the stories generated, we selected 3 pairs of stories from each story domain at random, with a gap of 1, 2, or 3 across the parameter upon which story generation was varied. Duplicate stories were prevented from appearing in the survey to ensure all stories presented to participants were novel. With the 15 pairs of stories selected, we produced a 5-point Likert-scale survey [15] to assess how participants felt about how the two stories in each pair compared. Users were kept blind to the scale upon which the stories were generated, and how they were generated to avoid biasing user responses.

For each question in the survey, the two paragraph-length stories within a given pair were presented as Story A and B. Following this, a statement was presented about the two stories, making an assertion concerning the domain-specific parameter relevant to their creation. For example, if a pair of stories was about two characters having a falling out, the statement might read: "Sarah and Emma's relationship deteriorates more in story A than in story B". Participants were then asked to respond to the statement on a 5-point Likert-Scale, indicating their level of agreement with that statement.

The presentation of Story A and B was randomized such that for some questions, Story A was intended to be more intense, or higher on the 1-10 scale relevant to its generation than Story B, and for others, Story B was intended to be more intense than story A. By randomizing which story was intended to be more intense, we control for potential order biases that might arise in user responses.

Examples of these questions and stories are given in the appendix.

After collecting responses, responses were reordered based on the intended intensity and the statement posed to the participants for that question. For example, if Story A was intended to be more intense and the statement users were responding to stated "A is more intense than B", a "Strongly Agree" response would be aligned with the intended intensity. Conversely, if Story B was intended to be more intense, a "Strongly Disagree" response would be aligned with the intended intensity.

As an additional measure, a single qualifying question was inserted, asking participants to respond to a statement about two stories with vastly different intended impact, both as assessed by the generative model and by the authors of this paper. This was included to disqualify all responses from participants who responded to this pair of stories improperly, as this would indicate a lack of care when responding to our survey. We additionally filtered out incomplete responses and responses completed in unrealistic time frames, which indicated skipping through the survey. 22 of the 28 responses were kept after filtering.

IV. Results

After collating responses and reordering them, we conducted a two-proportion z-test to determine if there were statistically significant differences in the proportions of users responding in certain ways to story pairings of prompt gaps 1, 2, and 3. These statistically significant inferences are displayed in Table 1, and data collected from the survey is displayed in Figures 1 and 2 below.

TABLE I
STATISTICALLY SIGNIFICANT DIFFERENCES IN USER RESPONSES ACROSS PROMPT GAPS

Comparison	z-value	p-value
Strongly Agree (Gap 1 $<$2)	-2.83	**0.0046**
Strongly Agree (Gap 1 $<$3)	-4.28	**$<$0.00001**
Disagree (Gap 1 $>$3)	2.76	**0.0028**
Disagree (Gap 2 $>$3)	2.57	**0.00508**
Neither Agree/Disagree (Gap 1 $>$3)	2.42	**0.0050**

As indicated in Table 1, we found that there was a statistically significant increase in the proportion of "Strongly Agree" responses from users between the "Prompt Gap 1" and "Prompt Gap 2" groups, as well as between the "Prompt Gap 1" and "Prompt Gap 3" groups. We also note a statistically significant decrease in the proportion of "Disagree" responses between the "Prompt Gap 1" and "Prompt Gap 3" groups, and that of the "Prompt Gap 2" and "Prompt Gap 3" groups. These changes in proportions of users responding in certain ways can be viewed in Fig. 1.

There is also a statistically significant decrease in the proportion of "Neither Agree or Disagree" responses between the "Prompt Gap 1" and "Prompt Gap 3" groups, trending downward from 18.2% at Prompt Gap 1 to 7.3% at Prompt Gap 3.

Overall, 78.5% of responses either agreed or strongly agreed with the intended ordering, with the proportion of those agreeing increasing alongside an increase in intended intensity gaps between pairs of stories.

There is additionally a downward trend in the amount of "Neither Agree or Disagree" responses as the prompt gap increases, from 18% with a prompt gap of 1, down to 7% with a prompt gap of 3. This can be viewed in Fig. 2.

We additionally examine the distributions of responses to individual questions in the survey, across our 5 story domains and prompt gap levels. These distributions are plotted in Fig. 3 through Fig. 5.

Notably, the majority of instances of disagreement are restricted to 3 of the 15 pairs of stories, 2 of which are within the same story domain, "Havoc Caused". While this may be a coincidence, it is worth noting that previous works [7] have identified a potential issue with safeguards OpenAI have integrated into their systems, noting a tendency of the model to lead to happy endings, forcing stories to resolve in an agreeable manner. It may be the case that such safeguards cause difficulties with story-generation tasks depicting harm or damage to people's lives.

The median response for questions of all prompt gaps was "Agree". The mode response for questions of prompt gap 1 and 2 was "Agree", while it was "Strongly Agree" for prompt gap 3. Generally, this indicates a strong alignment of user perception of these stories with their intended ordering.

V. Discussion

In line with our hypothesis, our results indicate that LLMs appear to be capable of generating distinguishable stories in response to parameterized prompts, aligning with author intentions. A significant majority of responses (78.5%) agree with the alignment of generated narratives with their intended intensity. This effect becomes more pronounced as the gap between compared stories widens, with statistically significant increases in the proportion of "Strongly Agree" responses and statistically significant decreases in the proportion of "Disagree" responses. Additionally, our data suggests that as the gap between compared stories increases, user certainty in their comparisons also rises. This is reflected in a decreasing proportion of users selecting "Neither Agree nor Disagree" as the intended prompt gap widens.

Another means of assessing the effectiveness of the LLM at producing narratives with perceptible intensity differences is examining the ratio of responses that "Agree" or "Strongly Agree" with the intended ordering, to those that "Disagree" or "Strongly Disagree". If users were equally likely to agree as they were to disagree, we would expect a distribution of user responses on the Likert-survey centered around "Neither Agree nor Disagree", resulting in an agreement-to-disagreement ratio of approximately 1:1.

Instead, for Prompt Gap 1, this ratio is 5.43. For Prompt Gap 2, this ratio is 7.64. For Prompt Gap 3, this ratio is a staggering 33. Even at relatively small differences in intended intensity, users are far more likely than not to perceive the difference in intensity as intended by the narrative generation process using GPT-4o. At a larger intended intensity gap, users were 33 times more likely to perceive the difference in intensity as intended, than not.

Overall, this indicates that LLMs appear to be largely effective at producing narratives with discernible intensity differences in response to parameterized prompting, suggesting competence in producing narrative discourse that meets authorial goals.

It's also worth noting that the majority of "Disagree" responses are concentrated within 3 of the 15 story pair questions. This may be indicative of these specific story domains being more difficult for our approach to generate distinguishable stories, but also could be the result of our low sample size coming into play. Further study may be needed to determine this.

VI. Conclusion

In this paper, we conduct a user study on the perception of parameterized LLM-generated narratives and find that the vast majority of user responses assess these generated narratives as aligning with their intended intensities or placement on the scale with which they were generated. This effect grows in strength as the gap between pairs of evaluated stories increases, increasing user certainty in their responses, and increasing the proportion of user responses that "Strongly Agree" with their intended ordering.

This is promising for storylet system approaches hoping to leverage LLMs in the development of interactive narrative games or tools, as it indicates that modern LLMs like GPT-4o are capable of tuning generated story fragments accurately to the prompts used to generate them. Therefore, one should be able to use these tools to generate prose that aligns with desired storylet post-conditions, if provided during prompting. Used in this context, LLMs could aid in easing the authorial burden of building such systems, increasing authorial leverage.

VII. Future Work

Though our survey makes clear that GPT-4o is generally capable of generating story fragments that align well with author intentions, our study is still limited in scope to GPT-4o, and does not investigate the capabilities of other commercially available LLM systems. Additionally, our study was conducted with a small body of participants, and conducting a large study with the same methodology could allow for more insights to be gleaned. For example, we observed that the majority of "Disagree" responses were concentrated in a small number of the questions users responded to. It may be the case that this is a result of the story generation methodology encountering more difficulties in generating distinguishable stories with specific story domains, but our volume of responses does not enable us to make an assertion on this backed by statistical

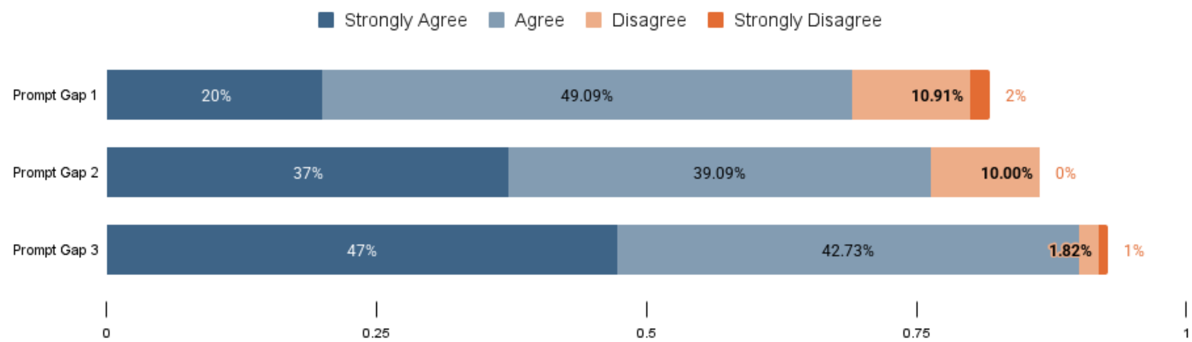


Fig. 1. User responses, adjusted to indicate their level of agreement with the intended intensity ordering of pairs of stories. Responses are grouped by the intended intensity gap between the pair of stories presented. There were 110 responses per prompt gap grouping.

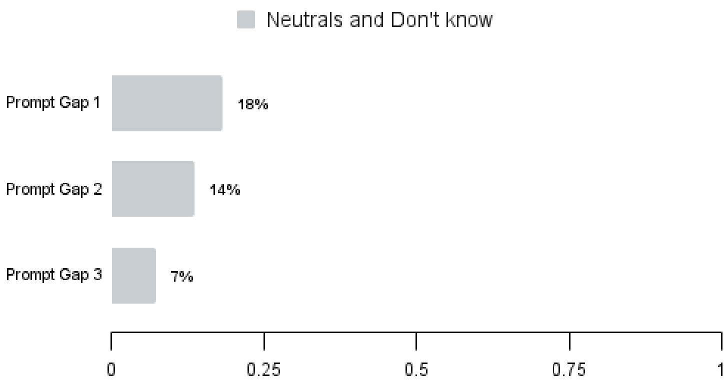


Fig. 2. Visualized is the proportion of user responses indicating they were unable to identify a significant difference between a given pair of stories. Responses are grouped by the intended intensity gap between the pair of stories presented.

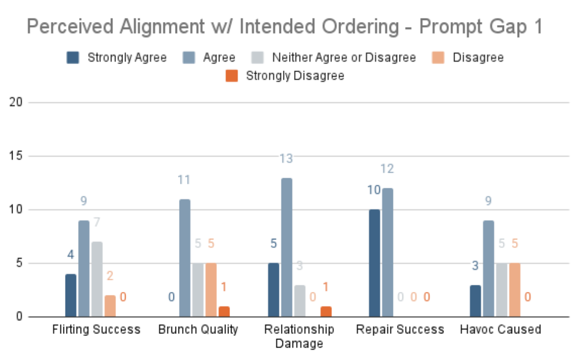


Fig. 3. The distribution of user responses on story pairs of gap 1. Response counts are present on the Y-axis, while story domain labels are present on the X-axis.

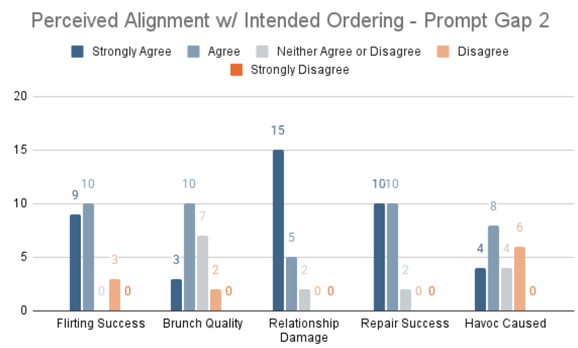


Fig. 4. The distribution of user responses on story pairs of gap 2.

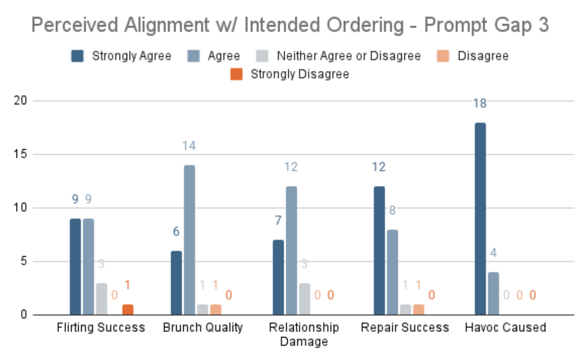


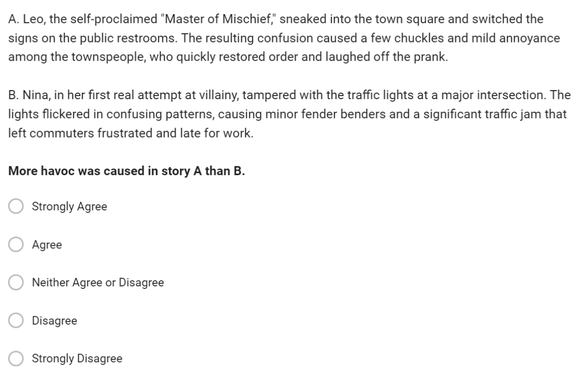
Fig. 5. The distribution of user responses on story pairs of gap 3. Notably fewer responses “Disagree” or “Strongly Disagree” with the intended ordering.

grounding. More work could be done expanding on our methodology to conduct a more thorough examination of this, as well as expanding the evaluation to other LLMs to see if some are more effective than others.

Additionally, We hope that in the future, studies will investigate the ability of LLMs to be integrated into hybrid LLM and rules-based systems built atop storylet-oriented narrative generation tools such as Lume [9] or College Ruled [10]. This study indicates that continuous advances in LLM technology are making these systems more viable for use in such contexts, and should be investigated further.

Appendix

Below we will provide a few example questions from the survey, to give a better picture of what the survey consisted of. Fig. 6 gives the example of the easiest pair of stories to distinguish in the survey, with 18 respondents strongly agreeing with the intended ordering, and the remaining 4 agreeing. Fig. 7 gives an example of an additional story domain, while Fig. 8 gives the example of what was the most difficult question for respondents to identify as intended, with 6 "Disagree" responses noted.



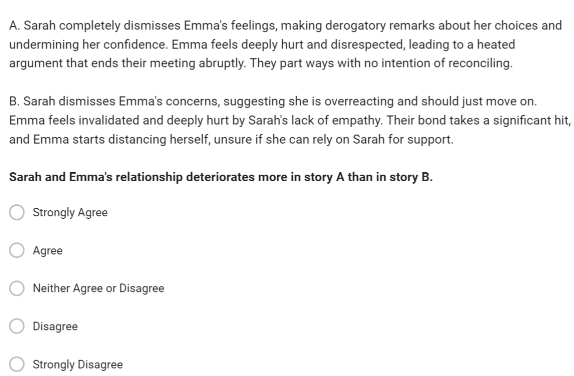
A. Leo, the self-proclaimed "Master of Mischief," sneaked into the town square and switched the signs on the public restrooms. The resulting confusion caused a few chuckles and mild annoyance among the townspeople, who quickly restored order and laughed off the prank.

B. Nina, in her first real attempt at villainy, tampered with the traffic lights at a major intersection. The lights flickered in confusing patterns, causing minor fender benders and a significant traffic jam that left commuters frustrated and late for work.

More havoc was caused in story A than B.

- Strongly Agree
- Agree
- Neither Agree or Disagree
- Disagree
- Strongly Disagree

Fig. 6. An example question as seen within the survey. The story domain is "Havoc Caused", with story A generated with an intended value of 1/10, and story B generated with an intended value of 4/10. A "Strongly Agree" response to this question would be logged as "Strongly Disagree"-ing with the intended ordering on the scale. The "Prompt Gap" for this question is 3, as the stories were generated with an intended gap of 3 points on the scale.



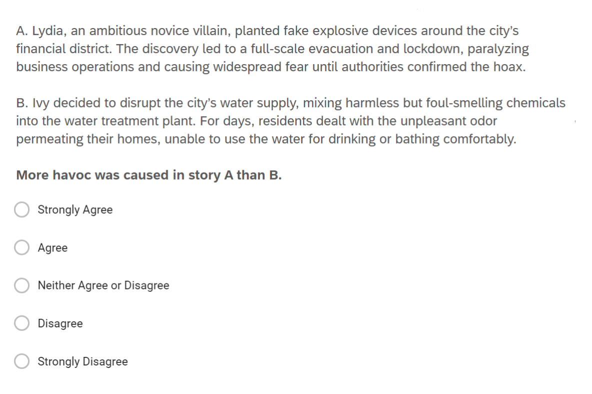
A. Sarah completely dismisses Emma's feelings, making derogatory remarks about her choices and undermining her confidence. Emma feels deeply hurt and disrespected, leading to a heated argument that ends their meeting abruptly. They part ways with no intention of reconciling.

B. Sarah dismisses Emma's concerns, suggesting she is overreacting and should just move on. Emma feels invalidated and deeply hurt by Sarah's lack of empathy. Their bond takes a significant hit, and Emma starts distancing herself, unsure if she can rely on Sarah for support.

Sarah and Emma's relationship deteriorates more in story A than in story B.

- Strongly Agree
- Agree
- Neither Agree or Disagree
- Disagree
- Strongly Disagree

Fig. 7. Another sample question. Story domain "Relationship Damage", with story A generated at a 10/10, and B generated at a 7/10.



A. Lydia, an ambitious novice villain, planted fake explosive devices around the city's financial district. The discovery led to a full-scale evacuation and lockdown, paralyzing business operations and causing widespread fear until authorities confirmed the hoax.

B. Ivy decided to disrupt the city's water supply, mixing harmless but foul-smelling chemicals into the water treatment plant. For days, residents dealt with the unpleasant odor permeating their homes, unable to use the water for drinking or bathing comfortably.

More havoc was caused in story A than B.

- Strongly Agree
- Agree
- Neither Agree or Disagree
- Disagree
- Strongly Disagree

Fig. 8. The most difficult question for respondents to identify as intended. Story domain "Havoc Caused", with story A generated at an 8/10, and B generated at a 6/10.

References

[1] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in neural information processing systems*, vol. 33, pp. 1877–1901, 2020.

[2] E. Short, "Storylets: You want them." Nov 2019. [Online]. Available: https://emshort.blog/2019/11/29/storylets-you-want-them/

[3] ——, "Beyond branching: Quality-based, salience-based, and waypoint narrative structures," Apr 2016. [Online]. Available: https://emshort.blog/2016/04/12/beyond-branching-quality-based-and-salience-based-narrative-structures/

[4] "Fallen london," 2009, browser-based interactive narrative game. [Online]. Available: http://fallenlondon.storynexus.com/

[5] S. Chen, M. Nelson, and M. Mateas, "Evaluating the authorial leverage of drama management," in *Proceedings of the AAAI conference on artificial intelligence and interactive digital entertainment*, vol. 5, no. 1, 2009, pp. 136–141.

[6] M. Kreminski and C. Martens, "Unmet creativity support needs in computationally supported creative writing," in *Proceedings of the First Workshop on Intelligent and Interactive Writing Assistants (In2Writing 2022)*, 2022, pp. 74–82.

[7] M. Dueifi and M. Eger, "The tomato festival: Towards using chatgpt for long-form discourse generation of plan-based narratives?" in *Proceedings of the AAAI conference on artificial intelligence and interactive digital entertainment*, 2024.

[8] M. Kreminski and N. Wardrip-Fruin, "Sketching a map of the storylets design space," in *International Conference on Interactive Digital Storytelling*. Springer, 2018, pp. 160–164.

[9] S. Mason, C. Stagg, and N. Wardrip-Fruin, "Lume: A system for procedural story generation," in *Proceedings of the 14th International Conference on the Foundations of Digital Games*, ser. FDG '19. New York, NY, USA: Association for Computing Machinery, 2019. [Online]. Available: https://doi.org/10.1145/3337722.3337759

[10] N. Treynor and J. McCoy, "College ruled: A pathfinding approach to generative storytelling," in *Proceedings of the 19th International Conference on the Foundations of Digital Games*, ser. FDG '24. New York, NY, USA: Association for Computing Machinery, 2024. [Online]. Available: https://doi.org/10.1145/3649921.3649994

[11] Y. Sun, P. J. Wang, J. J. Y. Chung, M. Roemmele, T. Kim, and M. Kreminski, "Drama llama: An llm-powered storylets framework for authorable responsiveness in interactive narrative," *arXiv preprint arXiv:2501.09099*, 2025.

[12] S. Ware and R. M. Young, "Glaive: a state-space narrative planner supporting intentionality and conflict," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 10, 2014, pp. 80–86.

[13] C. Chhun, F. M. Suchanek, and C. Clavel, "Do language models enjoy their own stories? prompting large language models for automatic story evaluation," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1122–1142, 2024.

[14] S. Shankar, J. Zamfirescu-Pereira, B. Hartmann, A. Parameswaran, and I. Arawjo, "Who validates the validators? aligning llm-assisted evaluation of llm outputs with human preferences," in *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, 2024, pp. 1–14.

[15] R. Likert, "A technique for the measurement of attitudes." *Archives of psychology*, 1932.